Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

There is a horizontal film of soap solution. On it a thread is placed in the form of a loop. The film is pierced inside the loop and the thread becomes a circular loop of radius R. If the surface tension of the loop be T, then tension in the thread will be:

A. $\pi R^2 T$

B. $2RT$

C. RT

D. $\frac{\pi R^2}{T}$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A particle executing SHM has a maximum speed of $0.5\ \mathrm{m\ s^{-1}}$ and a maximum acceleration of $1.0\ \mathrm{m\ s^{-2}}$. The angular frequency of oscillation is

A. $2\mathrm{rads}^{-1}$

B. $0.5\mathrm{rads}^{-1}$

C. $2\pi\mathrm{rads}^{-1}$

D. $0.5\pi\ \mathrm{rad}\ s^{-1}$

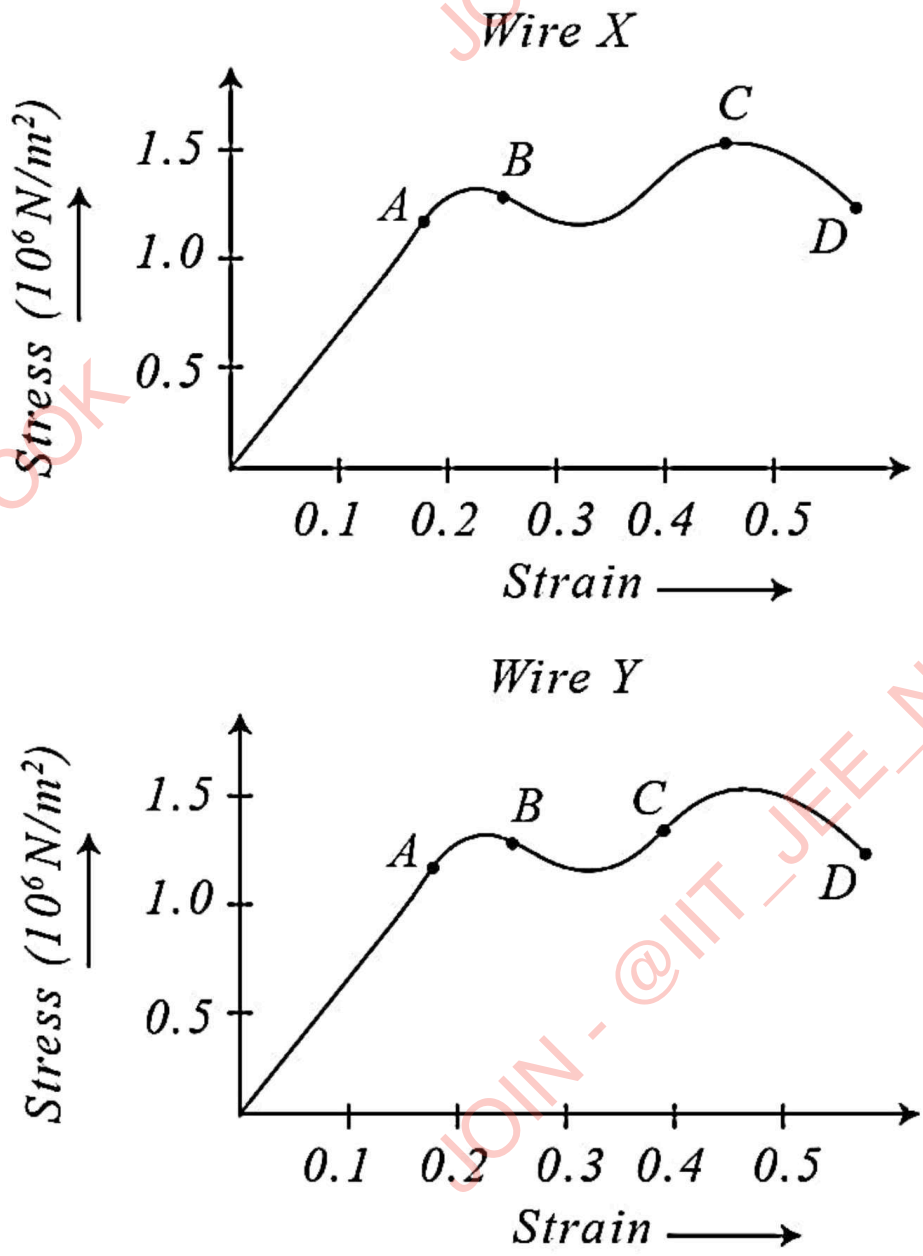
JOIN -@IIT_JEE_NEET_BOOK

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider two stress-strain graphs of two metal wires of different materials (say material X and Y)



Here, A, B, C and D represent the proportionality limit point, yield point, ultimate strength point and breaking point. Which of the following statement is correct?

A. Wire X is more ductile than wire Y.

B. Wire Y is more ductile than wire X.

C. Young's modulus of wire X is more than wire Y.

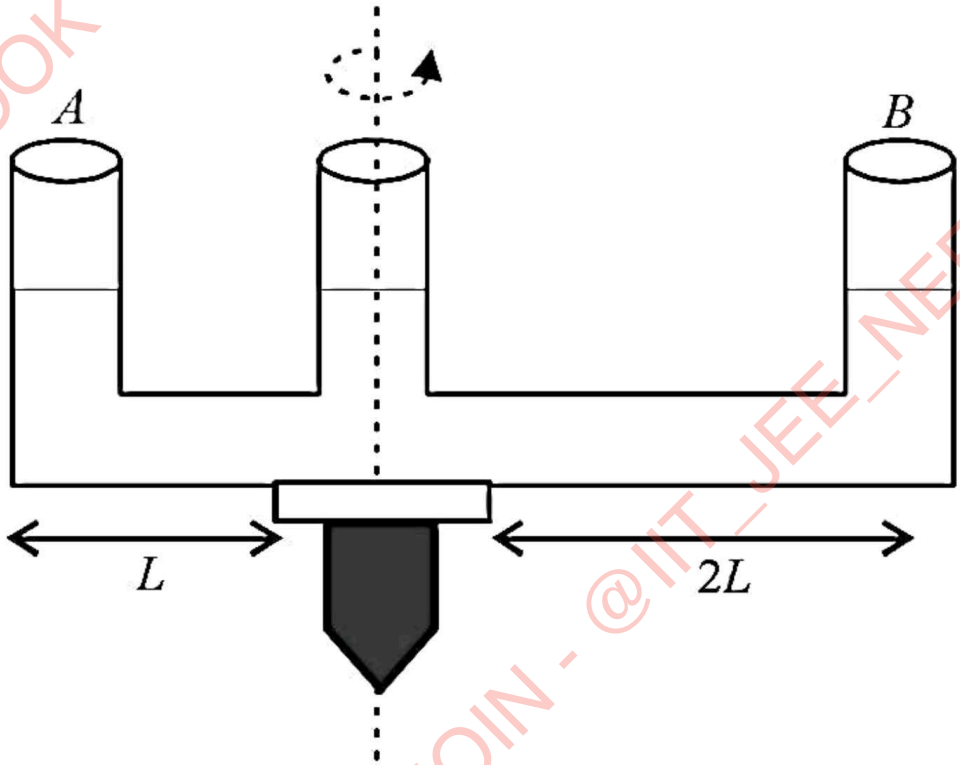
These wires cannot be compared as data is
D. insufficient.

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A given shaped glass tube having uniform cross section is filled with water and is mounted on a rotatable shaft as shown in figure. If the tube is rotated with a constant angular velocity ω when



A. water levels in both sections A and B go up

B. water level in section A goes up and that in B comes down

C. water level in section A comes down and that in B it goes up

D. water levels remain same in both sections.

JOIN - @IIT_JEE_NEET_BOOK

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Eight equal drops of water are falling through air with a steady velocity of $10\ \text{cms}^{-1}$. If the drops combine to form a single drop of big size, then the terminal velocity of this big drop is

A. $80\ \text{cms}^{-1}$

B. $30\ \text{cms}^{-1}$

C. $10\ \text{cms}^{-1}$

D. $40\ \text{cms}^{-1}$

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The position and velocity of a particle executing simple harmonic motion at $t = 0$ are given by 3 cm and 8 cm s^{-1} respectively. If the angular frequency of the particle is 2rads^{-1}, then the amplitude of oscillation (in cm) is

A. 3

B. 4

C. 5

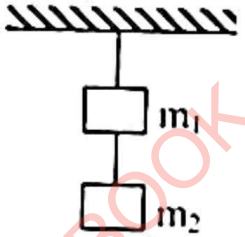
D. 6

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two wires of same material are hanging as shown in the figure. Breaking stress is equal for both the wires.



Assertion: If mass 'm_2' increases, then upper wire will break first.

Reason: Breaking load of two wires of same material can be different.

A. Assertion is true, reason is true; reason is a correct explanation for assertion.

B. Assertion is true, reason is true; reason is not a correct explanation for assertion.

C. Assertion is true, reason is false.

D. Assertion is false but, reason is true.

JOIN - @IIT_JEE_NEET_BOOK

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Water enters through end A with speed v_1 and leaves through end B with speed v_2 of a cylindrical tube AB. The tube is always completely filled with water. In case I, the tube is horizontal and in case II, it is vertical with end A upwards and in case III, it is vertical with end B upwards. We have $v_1 = v_2$ for

A. case I

B. case II

C. case III

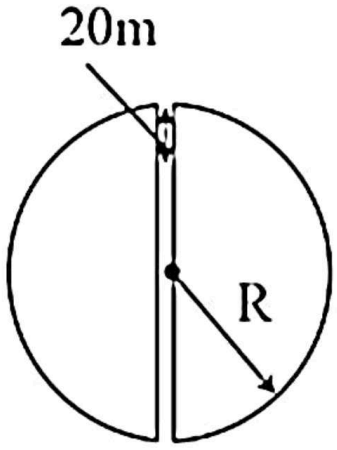
D. All cases

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A particle is dropped from depth of 20 m below the earth's surface along a tunnel created inside the planet of mass M with radius R as shown in the figure, the time period of oscillation is : (If $GM = 10R^3$, given $\pi^2 = 10$)



A. $4s$

B. 20 s

C. $2s$

D. $10s$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a cylinder-piston arrangement, the air is under a pressure P_1. A soap bubble of radius r lies inside the cylinder. Soap bubble has surface tension T. The radius of soap bubble is to be reduced to half. Find the pressure P_2 to which air should be compressed isothermally.

A. $P_1 + \frac{4T}{r}$

B. $4P_1 + \frac{12T}{r}$

C. $8P_1 + \frac{24T}{r}$

D. $P_1 + \frac{2T}{r}$

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The edge of an aluminium cube is $10\ \text{cm}$ long. One face of the cube is firmly fixed to a vertical wall. A mass of $100\ \text{kg}$ is then attached to the opposite face of the cube. The shear modulus of aluminium is 25GPa. What is the vertical deflection of this face?

A. $3.92 \times 10^{-7}\ \text{m}$

B. $5.98 \times 10^{-7}\ \text{m}$

C. $2.72 \times 10^{-7}\ \text{m}$

D. $4.82 \times 10^{-7}\ \text{m}$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A particle starts SHM from the mean position. Its amplitude is a and total energy E. At one instant its kinetic energy is $\frac{3E}{4}$. Its displacement at that instant is

A. $\frac{a}{\sqrt{2}}$

B. $\frac{a}{2}$

C. $\frac{a}{\sqrt{\left(\frac{3}{2}\right)}}$

D. $\frac{a}{\sqrt{3}}$

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The Young's modulus of steel is twice that of brass. Two wires of same length and of same area of cross-section, one of steel and another of brass are suspended from the same roof. If we want the lower ends of the wires to be at the same level, then the weights added to the steel and brass wires must be in the ratio of:

A. 2 : 1

B. 4 : 1

C. 1 : 1

D. 1 : 2

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The time period of a bob performing simple harmonic motion in water is 2 s. If density of bob is $\frac{4}{3} \times 10^3$ kg m^{-3}, then time period of bob performing simple harmonic motion in air will be

A. 3 s

B. 4 s

C. 2 s

D. 1 s

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A cylindrical tank is filled with water to level of 3 m. A hole is opened at height of 52.5 cm from bottom. The ratio of the area of the hole to that of cross-sectional area of the cylinder is 0.1 . The square of the speed with which water is coming out from the orifice is (Take $g = 10$ m s^{-2})

A. 50 m^2 s^{-2}

B. 40 m^2 s^{-2}

C. 51.5 m^2 s^{-2}

D. 50.5 m^2 s^{-2}

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The bob of a simple pendulum is a spherical hollow ball filled with water. A plugged hole near the bottom of the oscillating bob gets suddenly unplugged. During observation, till water is coming out, the time period of oscillation would

A. First increase and then decrease to the original value

B. First decrease and then increase to the original value

C. Remain unchanged

D. Increase towards a saturation value

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The bulk modulus of a spherical object is B. If it is subjected to uniform pressure p, the fractional decrease in radius is

A. $\frac{p}{B}$

B. $\frac{B}{3p}$

C. $\frac{3p}{B}$

D. $\frac{p}{3B}$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion : Smaller drops of liquid resist deforming forces better than the larger drops.
Reason : Excess pressure inside a drop is directly proportional to its surface area.

A. If both assertion and reason are true and reason is the correct explanation of assertion

B. If both assertion and reason are true but reason is not the correct explanation of assertion

C. If assertion is true but reason is false

D. If both assertion and reason are false.

JOIN - @IIT_JEE_NEET_BOOK

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The displacement of a particle is given at time t, by: $x = A\sin(-2\omega t) + B\sin^2\omega t$ Then,

A. the motion of the particle is SHM with an amplitude of $\sqrt{A^2 + \frac{B^2}{4}}$

B. the motion of the particle is not SHM, but oscillatory with a time period of $T = \pi/\omega$

C. the motion of the particle is oscillatory with a time period of $T = \pi/2\omega$

D. the motion of the particle is a periodic.

JOIN - @IIT_JEE_NEET_BOOK

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A solid sphere of volume V and density ρ floats at the interface of two immiscible liquids of densities ρ_1 and ρ_2 respectively. If $\rho_1 < \rho < \rho_2$, then the ratio of volume of the parts of the sphere in upper and lower liquid is

A. $\frac{\rho-\rho_1}{\rho_2-\rho}$

B. $\frac{\rho_2-\rho}{\rho-\rho_1}$

C. $\frac{\rho+\rho_1}{\rho+\rho_2}$

D. $\frac{\rho+\rho_2}{\rho+\rho_1}$

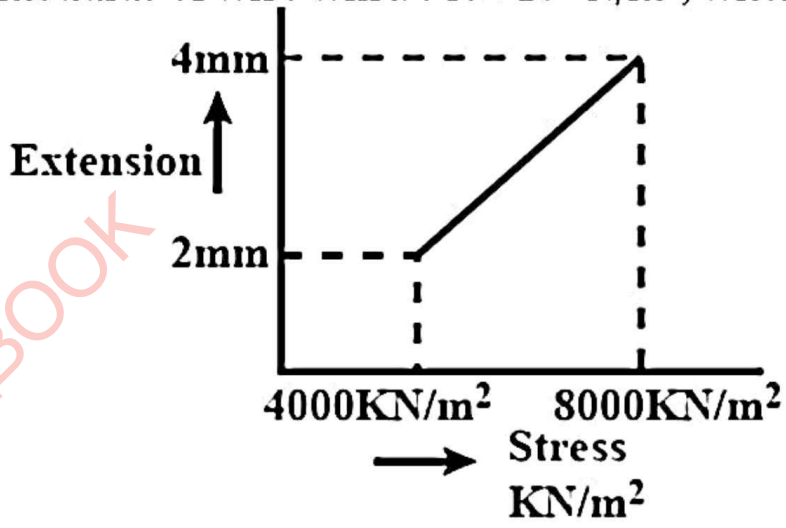
Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

In determination of young modulus of elasticity of wire, a force is applied and extension is recorded. Initial length of wire is '1m'. The curve between extension and stress is depicted then young modulus of wire will be K × 10^9 N/m^2, where K is

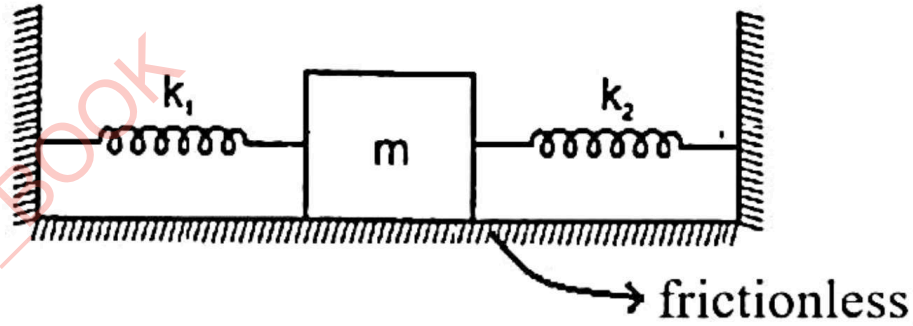


Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

Two springs with negligible masses and force constants $k_1 = 200\ \mathrm{N/m}$ and $k_2 = 160\ \mathrm{N/m}$ are attached to the block of mass $m = 10\ \mathrm{kg}$ as shown in figure. Initially the block is at rest, at the equilibrium position in which both springs are neither stretched nor compressed. At time $t = 0$, sharp impulse of 50 N-s is given to the block with a hammer along the spring. Then the amplitude of spring will be K metre. Find 6K



Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

A $0.1 - \mathrm{kg}$ mass is suspended from a wire of negligible mass. The length of the wire is $1\ \mathrm{m}$ and its cross-sectional area is $4.9 \times 10^{-7}\ \mathrm{m}^2$. If the mass is pulled a little in the vertically downward direction and released, it performs simple harmonic motion of angular frequency $140 \mathrm{rads}^{-1}$. If the Young modulus of the material of the wire is $n \times 10^9\ \mathrm{N\ m}^{-2}$, the value of n is

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

A sphere of mass M and radius R is falling in a viscous fluid. The terminal velocity attained by the falling object will be proportional to R^x, find x.

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

A rubber cord has a cross-sectional area $10^{-6}\ \mathrm{m}^2$ and total unstretched length $0.1\ \mathrm{m}$. It is stretched to $0.125\ \mathrm{m}$ and then released to project a particle of mass $5.0\ \mathrm{g}$. The velocity of projection is [Given, Young's modulus of rubber, $Y = 5 \times 10^8\ \mathrm{N\ m^{-2}}$]

Chemistry Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The experimental data for the reaction, $2\ \mathrm{A} + \mathrm{B_2} \rightarrow 2\mathrm{AB}$ is

Experiment	[A]	[$\mathrm{B_2}$]	Rate (mole $\mathrm{s^{-1}}$)
1	0.50	0.50	1.6×10^{-4}
2	0.50	1.00	3.2×10^{-4}
3	1.00	1.00	3.2×10^{-4}

The rate equation for the above data is

A. Rate $= \mathrm{k[A]^2[B]^2}$

B. Rate $= \mathrm{k[A]^2[B]}$

C. Rate $= \mathrm{k[B_2]}$

D. Rate $= \mathrm{k[B_2]^2}$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Statement - 1: Reverse osmosis is used to purify saline water.

Statement -2: Solvent molecules pass from concentrated to dilute solution through semipermeable membrane if high pressure is applied on solution.

A. Statement-1 is True, Statement - 2 is True; Statement - 2 is a correct explanation for Statement-1.

B. Statement - 1 is True, Statement is True; Statement - 2, is NOT a correct explanation for Statement - 1.

C. Statement - 1 is True, Statement - 2, is False.

D. Statement - 1 is False, Statement - 2 is True.

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Using the standard electrode potentials given below identify the correct statements from the following.

$$Fe^{2+} + 2e^- \longrightarrow Fe; E^\circ = -0.44\ V$$

$$Cu^{2+} + 2e^- \longrightarrow Cu; E^\circ = +0.34\ V$$

$$Ag^{+} + e^- \longrightarrow Ag; E^\circ = +0.80\ V$$

(i) Copper can displace iron from $FeSO_4$ solution.

(ii) Iron can displace copper from $CuSO_4$ solution.

(iii) Silver can displace copper from $CuSO_4$ solution.

(iv) Iron can displace silver from $AgNO_3$ solution.

JOIN - @IIT_JEE_NEET_BOOK

A. (i), (ii)

B. (ii), (iii)

C. (ii), (iv)

D. (i), (iv)

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The reaction, cis $Z \underset{K_b}{\overset{K_f}{\rightleftarrows}}$ trans Z is first order in both directions. At $28^{\circ}\mathrm{C}$, the equilibrium constant is 10^{-2} and the rate constant, $K_f = 3 \times 10^{-5}\ \mathrm{s}^{-1}$. In an experiment starting with the pure cis-form, how long would it take for half of the equilibrium amount of trans-isomer to be formed?

A. 5.6×10^4 s

B. 9.8×10^2 s

C. 2.3×10^2 s

D. 7.5×10^3 s

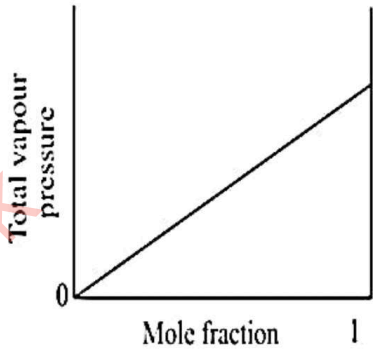
Question No. 5

Only One Option Correct Type

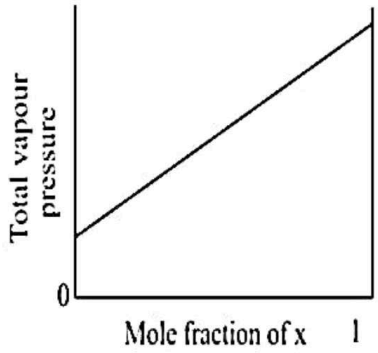
Each question has multiple options out of which ONLY ONE is correct.

The plot of total vapour pressure as a function of mole fraction of the components of an ideal solution formed by mixing liquids X and Y is:

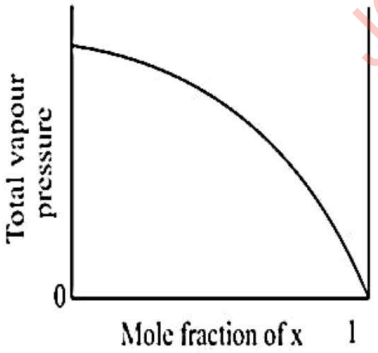
A.



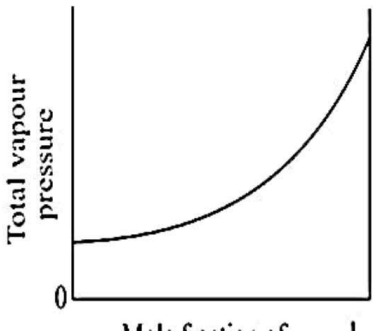
B.



C.



D.



Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If hydrogen and oxygen are mixed and kept in the same vessel at room temperature, the reaction does not take place to form water because

A. activation energy for the reaction is very high at room temperature

B. molecules have no proper orientation to react to form water

C. the frequency of collisions is not high enough for the reaction to take place

D. no catalyst is present in the reaction mixture.

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Suppose that gold is being plated on to another metal in an electrolytic cell. The half-cell reaction producing the Au(s) is $AuCl_4^- + 3e^- \rightarrow Au(s) + 4Cl^-$. If a 0.30 A current runs for 15.00 minute, what mass of Au(s) will be plated, assume all the electrons are used in the reduction of $AuCl_4^-$? The Faraday constant is 96485 C/mol and molar mass of Au is 197.

A. 0.184 g

B. 0.551 g

C. 1.84 g

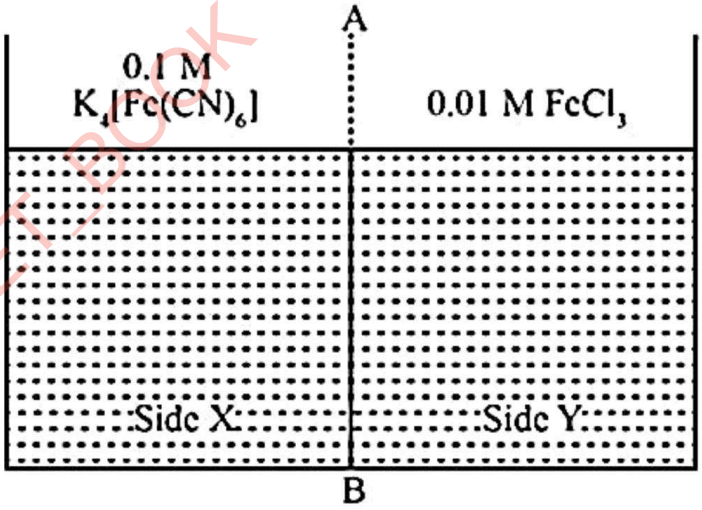
D. 0.613 g

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$FeCl_3$ on reaction with $K_4\left[Fe(CN)_6\right]$ in aqueous solution gives blue colour. These are separated by a semipermeable membrane AB as shown. Due to osmosis there is



A. blue colour formation in side X

B. blue colour formation in side Y

C. blue colour formation in both of the side X and Y

D. no blue colour formation

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For the electrochemical cell, $M|M^+||X^-|X$, $E^\circ\left(M^+ \mid M\right) = 0.44$ V and $E^\circ(X \mid X^-) = 0.33$ V. From this data one can deduce that

$M + X \rightarrow M^+ + X^-$ is the spontaneous

A. reaction

$M^+ + X^- \rightarrow M + X$ is the spontaneous

B. reaction

C. $E_{\text{cell}} = 0.77$ V

D. $E_{\text{cell}} = -0.77$ V

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The voltage of the cell consisting of Li (s) and $\mathrm{F}_2(\mathrm{~g})$ electrodes is 5.92 V at standard condition at 298 K. What is the voltage if the electrolyte consists of 2M LiF.

$\left(\ln 2 = 0.693, \mathrm{R} = 8.314 \mathrm{JK}^{-1}\ \mathrm{mol}^{-1}\right.$ and $\left.\mathrm{F} = 96500 \mathrm{Cmol}^{-1}\right)$

A. 5.90 V

B. 5.937 V

C. 5.88 V

D. 4.9 V

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the following statements,
The rate law for the acid catalysed hydrolysis of an ester being given as
$\text{Rate} = k[H^+][\text{ester}] = k'[\text{ester}]$.
If the acid concentration is doubled at constant ester concentration
1. The second order rate constant, k is doubled.
2. The pseudo first order rate constant, k is double.
3. The rate of the reaction is doubled.
Which of the above statements are correct?

A. 1 and 2

B. 2 and 3

C. 1 and 3

D. 1,2 and 3

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Boiling point of benzene is $353.23\ \text{K}$. When $1.8\ \text{g}$ of non-volatile solute is dissolved in $90\ \text{g}$ of benzene. Then boiling point is raised to $354.11\ \text{K}$. Given K_b (benzene) $= 2.53\ \text{kg mol}^{-1}$. The molecular mass of non-volatile substance is

A. $58\ \text{g mol}^{-1}$

B. $120\ \text{g mol}^{-1}$

C. $116\ \text{g mol}^{-1}$

D. $60\ \text{g mol}^{-1}$

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider this reaction :

$2\,NO_2\,(g) + O_3\,(g) \longrightarrow N_2O_5\,(g) + O_2\,(g)$

The rate law for the above reaction is given as rate $= k[NO_2][O_3]$. Which of the following mechanism is/are consistent with the rate law?

Mechanism I

$NO_2\,(g) + O_3\,(g) \xrightarrow{Slow} NO_3\,(g) + O_2\,(g)$

$NO_3\,(g) + NO_2\,(g) \xrightarrow{fast} N_2O_5\,(g)$

Mechanism II

$O_3\,(g) \rightleftarrows O_2\,(g) + O$

$NO_2\,(g) + O \xrightarrow{slow} NO_3\,(g)$

$NO_3\,(g) + NO_2\,(g) \xrightarrow{fast} N_2O_5\,(g)$

A. I only

B. II only

C. Both I & II

D. Neither I nor II

JOIN - @IIT_JEE_NEET_BOOK
u/wraphatass

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A 200 W, 100 V bulb is connected in series with an electrolytic cell. If an aqueous solution of an Sn-salt is electrolysed for 5hrs, 11.1 g of Sn gets deposited. The chemical formula of the compound is (Given atomic weight of Sn is 118.7 g mol^{-1}).

A. SnO

B. $SnCl_2$

C. $SnCl_4$

D. SnO_2

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If energy of activation of the reaction is 53.6 kJ mol^{-1} and the temperature changes from 27°C to 37°C then the value of $\left(\frac{k_{37^\circ C}}{k_{27^\circ C}}\right)$ is

A. 2.5

B. 1.0

C. 2.0

D. 1.5

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The conductivity of a weak acid HA of concentration $0.001\ \text{mol}\ L^{-1}$ is $2.0 \times 10^{-5}\ \text{S}\ \text{cm}^{-1}$. If $\Lambda_m^0\left(HA\right) = 190\ \text{S}\ \text{cm}^2\ \text{mol}^{-1}$, the ionization constant (K_a) of HA is equal to $____ \times 10^{-6}$
(Round off to the Nearest Integer)

A. 11

B. 13

C. 12

D. 14

JOIN - @IIT_JEE_NEET_BOOK

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion : Azeotropic mixtures are formed only by non-ideal solutions and they may have boiling points either greater than both the components or lesser than both the components.
Reason: The composition of the vapour phase is same as that of the liquid phase of an azeotropic mixture.

A. If both assertion and reason are true and reason is the correct explanation of assertion.

B. If both assertion and reason are true but reason is not the correct explanation of assertion.

C. If assertion is true but reason is false.

D. If both assertion and reason are false.

JOIN - @IIT_JEE_NEET_BOOK

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A 1.0M solution with respect to each of the metal halides AX_3, BX_2, CX_3 and DX_2 is electrolysed using platinum electrodes. If $E^{\circ}_{A^{3+}/A} = 1.50\ V, E^{\circ}_{B^{2+}/B} = 0.3\ V, E^{\circ}_{C^{3+}/C} = -0.74\ V,$ $E^{\circ}_{D^{2+}/D} = -2.37\ V$. The correct sequence in which the various metals are deposited at the cathode is

A. A, B, C, D

B. A, B, C

C. D, C, B, A

D. C, B, A

JOIN - @IIT_JEE_NEET_BOOK
@wrapthatass

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Regarding to Henry's law which of the following is not correct statement ?

A. Different gases have different K_H (Henry's law constant) values at the same temperature

B. The partial pressure of the gas in vapour phase is proportional to the mole fraction of the gas in the solution

C. Higher the value of K_H at a given pressure, higher is the solubility of the gas in the liquids

D. The value of K_H increases with increase of temperature and K_H is function of the nature of the gas

JOIN - @IIT_JEE_NEET_BOOK

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Mark the correct statement(s) for corrosion.

A. It quickly coats the surfaces of the metallic object.

B. Its examples are- Rusting of iron, tarnishing of silver

C. It does not cause damage to buildings, ship made metals especially that are made up of iron.

D. All of the above

Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

If the freezing point of a 0.01 molal aqueous solution of a cobalt (III) bromide-ammonia complex (which behaves as a strong electrolyte) is $-0.0557^{\circ}\mathrm{C}$, the number of bromide(s) in the coordination sphere of the complex is [$\mathrm{K_f}$ for water $= 1.86\ \mathrm{K\ kg\ mol^{-1}}$]

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

Standard free energies of formation (in $\mathrm{kJ/mol}$) at $298\ \mathrm{K}$ are -237.2, -394.4 and -8.2 for $\mathrm{H_2O(l)}$, $\mathrm{CO_2(g)}$ and pentane (g), respectively. The value $\mathrm{E^{\circ}_{cell}}$ for the pentane-oxygen fuel cell is (mark answer to nearest integer)

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

A and B decompose via first order kinetics with half-lives 54.0 min and 18.0 min respectively. Starting from an equimolar non-reactive mixture of A and B, the time taken for the concentration of A to become 16 times that of B is _____ min. (Round off to the Nearest Integer).

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

The forward rate constant for the reversible gaseous reaction, $C_2H_6 \rightleftharpoons 2CH_3$ is $3.14 \times 10^2\ s^{-1}$ at 200 K. If 10^{-5} moles of CH_3 and 100 mol of C_2H_6 are present in 10 L vessel at equilibrium, then the rate constant for the backward reaction at this temperature is ..... $10^{15}\ L\ mol^{-1}\ s^{-1}$ (mark answer to nearest integer)

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

A current of 2.0 A when passed for 5 hours through a molten metal salt deposits 22.2 g of metal of atomic weight 177 . The oxidation state of the metal in the metal salt is

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A point P moves on xy −plane such that $PS + PS' = 4$, where $S(K,\ 0)$ and $S'(-K,\ 0)$, then which of the following is not true about the locus of P ?

A. Ellipse if $K \in (-2,\ 2)$

B. Line segment if $K = \pm 2$

C. Empty if $K \in (-\infty,\ -2) \cup (2,\ \infty)$

D. None of these

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

OA and OB are two mutually perpendicular chords of $y^2 = 4ax$, O being the origin. Line AB will always passes through the point

A. $(6a, 0)$

B. $(8a, 0)$

C. $(4a, 0)$

D. $(a,\ 0)$

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the three distinct lines $x + 2ay + a = 0, x + 3by + b = 0$ and $x + 4ay + a = 0$ are concurrent, then the point (a, b) lies on a

A. circle

B. straight line

C. parabola

D. hyperbola

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If a variable line $3x + 4y - \lambda = 0$ is such that the two circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 18x - 2y + 78 = 0$ are on its opposite sides, then the set of all values of λ is the interval :

A. $[13, 23]$

B. $(23, 31)$

C. $[12, 21]$

D. $(2, 17)$

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of rational point(s) (a point $(a,\ b)$ is rational, if a and b both are rational numbers) on the circumference of a circle having centre (π,e) is

A. at most one

B. at least two

C. exactly two

D. infinite

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If two points P and Q lie on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1\left(a<b\right)$, whose centre C be such that CP is perpendicular to CQ, then the value of $\frac{1}{CP^2}+\frac{1}{CQ^2}$ is

A. $\frac{b^2-a^2}{2ab}$

B. $\frac{1}{a^2}+\frac{1}{b^2}$

C. $\frac{2ab}{b^2-a^2}$

D. $\frac{1}{a^2}-\frac{1}{b^2}$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A straight line L through the point (3, -2) is inclined at an angle 60° to the line $\sqrt{3}x + y = 1$. If L also intersects the x-axis, then the equation of L is

A. $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$

B. $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$

C. $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$

D. $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The points $A(3, 6)$ and B lie on the parabola $y^2 = 4ax$, such that the chord AB subtends 90° at the origin, then the length of the chord AB is equal to

A. $15\sqrt{13}$ units

B. $12\sqrt{17}$ units

C. $9\sqrt{17}$ units

D. $9\sqrt{10}$ units

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Suppose S and $S\prime$ are foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$. If P is a variable point on the ellipse and if Δ is the area (in sq. units) of the triangle $PSS\prime$, then the maximum value of Δ is double of

A. Minimum value of $\frac{2a^8+2b^4}{a^4b^2}\ \forall a,\ b\in R$

B. Minimum value of $\frac{3a^8+3b^4}{a^4b^2}\ \forall a,\ b\in R$

C. $\frac{4a^8+4b^4}{a^4b^2}\ \forall a,\ b\in R$

D. $\frac{6a^8+6b^4}{a^4b^2}\ \forall a,\ b\in R$

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two straight lines having variable slopes m_1 and m_2 pass through the fixed points $(a,0)$ and $(-a,0)$ respectively. If $m_1m_2=2$, then the eccentricity of the locus of the point of intersection of the lines is

A. $\sqrt{2}$

B. $\sqrt{3}$

C. 2

D. $\sqrt{\frac{3}{2}}$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let L be the line passing through the point $P(1, 2)$ such that its intercepted segment between the co-ordinate axes is bisected at P. If L_1 is the line perpendicular to L and passing through the point $(-2,\ 1)$, then the point of intersection of L and L_1 is

A. $\left(\frac{3}{5}, \frac{23}{10}\right)$

B. $\left(\frac{4}{5}, \frac{12}{5}\right)$

C. $\left(\frac{11}{20}, \frac{29}{10}\right)$

D. $\left(\frac{3}{10}, \frac{17}{5}\right)$

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let (x, y) be a variable point on the curve $4x^2 + 9y^2 - 8x - 36y + 15 = 0$. Then $\min\left(x^2 - 2x + y^2 - 4y + 5\right) + \max\left(x^2 - 2x + y^2 - 4y + 5\right)$ is -

A. $\frac{325}{36}$

B. $\frac{36}{325}$

C. $\frac{13}{25}$

D. $\frac{25}{13}$

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The locus of the point (x, y) whose distance from the line $y = 2x + 2$ is equal to the distance from (2,0), is a parabola with the length of latus rectum same as that of the parabola $y = Kx^2$, then the value of K is equal to

A. $\frac{\sqrt{5}}{12}$

B. $\frac{\sqrt{5}}{4}$

C. $\frac{4}{\sqrt{5}}$

D. $\frac{12}{\sqrt{5}}$

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The vertices of the triangle ABC are $(2,\ 1),\ (4,\ 3)$ and $(2,\ 5)$. $D,\ E,\ F$ are the mid-points of the sides. The area of the triangle DEF is

A. 1

B. 1.5

C. 3

D. 4

JOIN - @IIT_JEE_NEET_BOOK

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The equation of a circle with origin as centre and passing through the vertices of an equilateral triangle whose median is of length $3a$ is

A. $x^2 + y^2 = 9a^2$

B. $x^2 + y^2 = 16a^2$

C. $x^2 + y^2 = 4a^2$

D. $x^2 + y^2 = a^2$

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Suppose the parabola $(y-k)^2 = 4(x-h)$, with vertex A, passes through $O = (0,0)$ and $L = (0,2)$. Let D be an end point of the latus rectum. Let the y-axis intersect the axis of the parabola at P. Then $\angle PDA$ is equal to

A. $\tan^{-1}\frac{1}{19}$

B. $\tan^{-1}\frac{2}{19}$

C. $\tan^{-1}\frac{4}{19}$

D. $\tan^{-1}\frac{8}{19}$

JOIN - @IIT_JEE_NEET_BOOK
u/wrapthatass

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

On a rectangular hyperbola $x^2 - y^2 = a^2, a > 0$, three points $A,\ B,\ C$ are taken as follows : $A = (-a,\ 0)$; B and C are placed symmetrically with respect to the X-axis on the branch of the hyperbola not containing A. Suppose that the ΔABC is equilateral. If the side length of the ΔABC is ka, then k lies in the interval

A. $(0, 2]$

B. $(2, 4]$

C. $(4, 6]$

D. $(6, 8]$

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The point $P(2, 1)$ is shifted parallel to the line $x + y = 1$, by a distance of $5\sqrt{2}$ in the direction of increasing ordinate, to reach the point Q. The image of Q in the line $y = -x$ is

A. (-3, 6)

B. (6, 3)

C. $(-4, -7)$

D. (6, -3)

JOIN - @IIT_JEE_NEET_BOOK

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Length of the focal chord of the parabola $(y+3)^2 = -8(x-1)$, which lies at a distance of 2 units from the vertex of the parabola, is

A. $8.$

B. $6\sqrt{2}.$

C. $9.$

D. $5\sqrt{3}.$

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The orthocentre of the triangle with vertices $(-2,\ -6),\ (-2,\ 4)$ and $(1,\ 3)$ is

A. $(-3,\ 1)$

B. $(-1, 1/3)$

C. $(1,\ 3)$

D. None of these

JOIN - @IIT_JEE_NEET_BOOK

Mathematics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

Let C_1 be the graph of $xy = 1$ and the reflection of C_1 in the line $y = 2x$ is C_2. If the equation of C_2 is expressed as $12x^2 + bxy + cy^2 + d = 0$, then the value of $(b + c + d)$ is equal to

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

The vertices of the triangle ABC are $A(0,0)$, $B(3, 0)$ and $C(3, 4)$, where A and C are foci of an ellipse and B lies on the ellipse. If the length of the latus rectum of the ellipse is $\frac{12}{p}$ units, then the value of 2p is

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

Let in ΔABC coordinates of vertex A is $(0,\ 0)$. Equation of the internal angle bisector of $\angle ABC$ is $x + y - 1 = 0$ and mid-point of BC is $(1,\ 3)$. The ordinate of vertex C is

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

Let PQ be the focal chord of the parabola $\mathrm{y}^2 = 4\mathrm{x}$. If the centre of the circle having PQ as its diameter lies on the line $\mathrm{y} = \frac{-4}{\sqrt{5}}$, then the length (in units) of chord PQ is M, find 10 M

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

If the circle $(x-a)^2 + y^2 = 25$ intersects the circle $x^2 + (y-b)^2 = 16$ in such a way that the length of the common chord is 8 units, then the value of $a^2 + b^2$ is

JOIN - @IIT_JEE_NEET_BOOK